JASA ARTICLE

Automated detection of fin whales with distributed acoustic sensing in the Arctic and Mediterranean

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ABSTRACT:
The fin whale is a key species in marine ecosystems and a sensitive indicator of ocean health. Yet monitoring its low-frequency calls at scale remains challenging. Distributed acoustic sensing (DAS) on subsea fiber cables can record thousands of calls daily, but these are hidden in terabytes of data, demanding automated detection. This study presents a three-stage pipeline: (i) data enhancement using frequency-wavenumber and root-mean-square filtering; (ii) call detection in the time-space domain using four approaches: line-based detection (Hough transform, HT), density-based clustering (DBSCAN), template matching combined with clustering (TM), and a deep-learning-based detector (YOLO); and (iii) detection refinement using hyperbola fitting and a LightGBM classifier. The pipeline is evaluated on datasets from two submarine cables: 135 km in Svalbard, Norway, and 162 km between Monaco and Italy. YOLO achieves the highest F1 score (0.89) on the Svalbard dataset, outperforming HT (0.57), DBSCAN (0.43), and TM (0.71). Without fine-tuning, YOLO also achieves a high F1 score (0.80) on the Monaco–Italy dataset, demonstrating robust generalization across geographic locations and seafloor environments. These results highlight DAS combined with deep learning as a powerful tool for scalable, real-time monitoring of marine mammal vocalizations. © *2025 Acoustical Society of America.* https://doi.org/10.1121/10.0041855

(Received 2 July 2025; revised 29 October 2025; accepted 20 November 2025; published online 29 December 2025)

[Editor: Shima Abadi] Pages: 5057–5074

I. INTRODUCTION

The fin whale (*Balaenoptera physalus*), the second-largest whale species, plays a crucial role in marine ecosystems both as a top consumer and a key contributor to ocean nutrient cycling. By feeding on krill and small fish, fin whales help regulate prey populations and enhance ocean productivity through nutrient redistribution (Aguilar and García-Vernet, 2018). Their widespread presence in temperate and polar waters makes them an important indicator of ocean health, as changes in their distribution reflect shifts in climate and prey availability. Despite past commercial whaling, some populations have started to recover, yet threats such as ship collisions and underwater noise remain (Edwards *et al.*, 2015). Effective detection and monitoring of fin whales are essential not only for their conservation but also for understanding broader environmental changes in the oceans.

Several methods have been developed to monitor and localize fin whales. Visual surveys, including ship-based and aerial observations, have traditionally been used to estimate populations and study their behavior (Watkins, 1981). Optical imaging from unmanned aerial vehicles (UAVs), such as drones, has improved detection efficiency, but coverage is restricted to specific survey transects. Satellite imagery provides broader coverage, especially in regions with limited human access (Mizroch *et al.*, 2009), but is still sparsely sampled. Tagging methods, including satellite and radio telemetry, provide precise data on whale movements, revealing seasonal migration patterns (Lydersen *et al.*, 2020), though tags often detach and can be applied to only a limited number of individuals. Passive acoustic monitoring (PAM) is widely used to record whale vocalizations, providing insights into their presence and movement patterns, but it typically relies on stationary instruments that are costly to maintain and require frequent retrieval for data analysis. Long-term acoustic networks and hydrophone arrays have been successfully applied for fin whale population studies (McDonald and Fox, 1999; Wilcock, 2012), but these systems also face challenges such as sparse spatial distribution and logistical constraints in deep-sea environments. More recently, environmental DNA sampling is an emerging tool that detects whale presence by analyzing genetic material in seawater (Alter *et al.*, 2022; Rodriguez *et al.*, 2025), offering a non-invasive alternative for population monitoring, but it is highly sensitive to environmental conditions and lacks the temporal resolution to track real-time movements of whales. While these methods together contribute to a comprehensive understanding of fin whale distribution and migration, their limitations highlight the need for complementary or improved monitoring approaches.

In recent years, advances in fiber-optic sensing have introduced distributed acoustic sensing (DAS) as a novel

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tool for whale monitoring (Landrø *et al.*, 2022; Rørstadbotnen *et al.*, 2023; Xiao *et al.*, 2025). DAS utilizes existing submarine fiber-optic cables as dense acoustic sensor arrays, detecting underwater sound waves, including whale vocalizations. DAS works by sending repeated laser pulses through the fiber-optic cable and measuring phase rate changes in the backscattered light due to the external wavefield. These phase rate shifts are linearly related to axial strain along the fiber. The strain data enables one to reconstruct parts of the acoustic environment. Unlike traditional PAM, which relies on dedicated hydrophone deployment, DAS offers continuous, real-time monitoring over vast areas without additional sensor infrastructure, making it a cost-effective and scalable approach for whale tracking in the deep seas and other remote environments.

Bouffaut *et al.* (2022) and Rørstadbotnen *et al.* (2023) used DAS to monitor whale calls in Svalbard, Norway. Bouffaut *et al.* (2022) focused on detecting and analyzing baleen whale calls along a 120 km-long fiber optic cable, utilizing spectrogram analysis, frequency filtering, and spatio-spectral representations to classify fin and blue whale vocalizations. While demonstrating DAS's potential for continuous monitoring, their method relied on manual annotation, limiting scalability for long-term monitoring. Rørstadbotnen *et al.* (2023) expanded on this approach by analyzing DAS data from two fiber cables and implementing two localization methods - a grid search optimization and a Bayesian filter. Their approach successfully estimated multiple whale positions and movement patterns along the two parallel 260 km-long fiber-optic cables. Similarly, Wilcock *et al.* (2023) analyzed four days of DAS data for studying fin and blue whale vocalizations from Oregon, USA. They used bandpass filtering and frequency-wavenumber (FK) filtering to enhance detection. Whale calls were manually identified, and the time difference of arrival was applied for localization. More recently, Goestchel *et al.* (2025) enhanced fin whale vocalizations using hybrid filtering, matched filtering, and noise envelope subtraction. Their work primarily focused on signal enhancement and denoising, rather than an end-to-end automated detection framework. Across these studies, call identification and time-of-arrival picking remained manual, limiting real-time monitoring at large spatial and temporal scales.

In this study, we aim to address this limitation of manual identification in massive data sizes by developing and systematically evaluating automated detection methods for fin whale monitoring using DAS. Unlike previous studies that relied on manual annotation or focused on specific detection techniques in isolation, we provide a comparative assessment of different automated approaches. Our method is tested using the same DAS data collected in Svalbard, Norway, as used by Bouffaut *et al.* (2022) and Rørstadbotnen *et al.* (2023), and a new DAS dataset in Monaco–Italy. By automating detection, we seek to enhance the scalability and efficiency of DAS-based monitoring, reducing reliance on manual annotation, enabling real-time tracking of fin whale populations and decreasing the big data load associated with DAS data acquisition.

The structure of this paper is as follows: Section II provides an overview of the DAS system and the datasets collected in Svalbard and Monaco–Italy. Section III outlines the detection pipeline: preprocessing, detection with the Hough transform (HT), Density-based spatial clustering of applications with noise (DBSCAN), template matching (TM), followed by clustering, and You only look once (YOLO); and refinement using hyperbola fitting and light gradient boosting machine (LightGBM). Section IV evaluates the detection performance across the Svalbard and Monaco–Italy datasets. Section V describes the real-time data processing pipeline, including batch processing, storage, and visualization. Finally, Sec. VI summarizes the main findings and explores future directions for fin whale monitoring using DAS.

II. DAS SYSTEMS AND DATASETS

A. DAS systems

DAS is a technology that leverages conventional telecommunication optical fibers to form dense linear arrays of virtual sensors, enabling the detection of strain or acoustic disturbances continuously along the fiber length (Taweesintananon *et al.*, 2021; Landrø *et al.*, 2022). A DAS interrogator sends laser pulses at a fixed pulse repetition frequency, f_s, which also defines the system's temporal sampling rate, into the fiber. As pulses propagate, a fraction of the light is backscattered continuously, primarily by Rayleigh scattering. By measuring the travel time of these backscattered signals, the interrogator determines the location of acoustic disturbances along the fiber.

To avoid overlap of backscattered signals, the pulse repetition rate must satisfy

$$f_s \leq \frac{c_{\text{fiber}}}{2L}, \tag{1}$$

where L is the total fiber length and c_{fiber} is the pulse propagation velocity in the fiber. This velocity is equal to the speed of light in vacuum ($c \approx 3 \times 10^8$ m/s) divided by the fiber's refractive index (n), i.e., $c_{\text{fiber}} = c/\text{n}$. For standard single-mode fibers, with a refractive index typically around 1.468, this results in c_{fiber} being approximately $2 \times 10^8\,\text{m/s}$. For example, for a fiber length of 150 km, the maximum pulse repetition rate is approximately 667 Hz.

Modern DAS systems primarily employ phase-sensitive coherent optical reflectometry techniques, where the interferometer measures the differential phase of Rayleigh backscatter between successive gauge-length sections. The resulting optical phase change $\Delta\phi$ due to an axial strain η over a gauge length L_g is given by

$$\Delta\phi = \frac{4\pi \text{n}\, \xi L_g}{\lambda}\, \eta, \tag{2}$$

where $\xi \approx 0.79$ is the strain-optic coefficient, and λ is the laser wavelength (Waagaard *et al.*, 2021).

Unlike standard DAS systems that transmit short, single-frequency laser pulses, the OptoDAS system from Alcatel Submarine Networks (ASN) used in this study employs frequency sweep interrogation (FSI), which transmits frequency-modulated pulses with durations typically around 100 μs. While a pulse of this 100 μs duration would physically occupy approximately 20 000 m in the fiber (given a pulse propagation velocity of 2×10^8 m/s), the spatial resolution of FSI-based DAS is actually meter-scale (Waagaard *et al.*, 2021). This high resolution is achieved because spatial resolution in FSI is determined by the frequency sweep bandwidth and subsequent pulse compression techniques, rather than the raw pulse duration. This approach of using longer, frequency-modulated pulses also has the additional benefit of significantly increasing the optical energy input, hence facilitating longer sensing lengths exceeding 100 km (Waagaard *et al.*, 2021).

The gauge length L_g in OptoDAS is software-defined and affects the sensitivity and data quality by controlling spatial averaging. Short gauge lengths are necessary to sample short-wavelength acoustic signals, while longer gauge lengths can enhance the signal-to-noise ratio (SNR) through increased averaging. A typical practical value balancing spatial resolution and sufficient SNR is 10 m (Waagaard *et al.*, 2021).

Overall, DAS enables real-time, cost-effective acoustic monitoring over extensive ocean regions, leveraging existing submarine cables for scalable marine mammal detection.

B. Svalbard data

We use DAS data collected along an existing submarine telecommunication cable running between Longyearbyen and Ny-Ålesund in Svalbard, Norway [Figs. 1(a) and 1(c)]. The cable contains standard single-mode G.652D fibers and is installed 0–2 m below the seafloor (ITU-T, 2024). It is roughly 250 km long and contains 24 individual fibers. We use one of these dark fibers for DAS measurements. The fiber is connected to the interrogator only at one end and left open at the far end, with no reflective termination, so the backscatter signal naturally fades with distance. Due to optical loss (0.18 dB/km), the backscattered signal falls below the noise floor after 135 km, so only the first 135 km are used for analysis. The cable is owned and operated by SIKT—Norway's National Education and Research Network (Landrø *et al.*, 2022; Taweesintananon *et al.*, 2023; Rørstadbotnen *et al.*, 2023).

For the DAS measurements, an ASN OptoDAS interrogator was deployed at onshore facilities in Longyearbyen

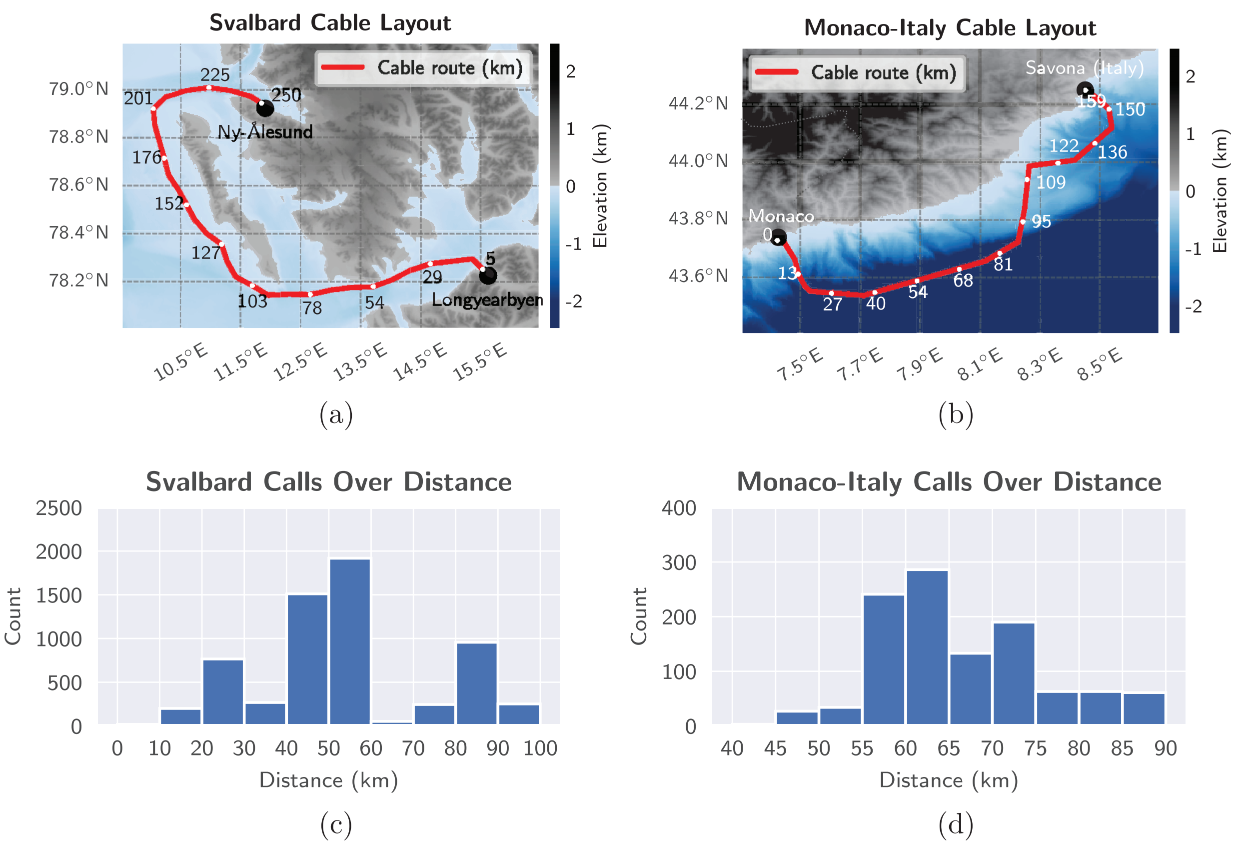

FIG. 1. Submarine cable layouts and fin whale vocalization distributions. (a) Fiber-optic cable between Longyearbyen and Ny-Ålesund, Svalbard, Norway. Adapted from Rørstadbotnen *et al.* (2023). "Simultaneous tracking of multiple whales using two fiber-optic cables in the Arctic," Front. Mar. Sci. **10**, 1130898; licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license (https://creativecommons.org/licenses/by/4.0/). (b) Fiber-optic cable between Monte Carlo, Monaco, and Savona, Italy (Migeon, 2006). (c) For the Svalbard cable, 12 h of data from 22 August 2022 show that most fin whale vocalizations occur between 20 km and 60 km along the fiber from the interrogator. (d) For the Monaco–Italy cable, 3 h of data from 29 January 2025 show that most vocalizations occur between 55 and 75 km.

and Ny-Ålesund during the summer of 2022. The interrogator sent linear, frequency-modulated optical pulses into the fiber, centered at a free-space wavelength of 1550 nm with a sweep bandwidth of approximately 80 MHz. For this study, we focus on 12 h of data from 22 August 2022, recorded within this 135 km section at a 625 Hz temporal sampling rate. The signals are sampled every 4.08 m with a gauge length of 8.16 m.

The SNR decreases at the far end of the cable due to optical loss, whereas near the shore, external vibrations associated with human activity can mask the fin whale calls. More specifically, over the first 70–80 km, the optical noise level is flat, and the background is dominated by external sources, with only small contributions from optical noise. Beyond this distance, optical noise increases and gradually dominates the background. Figure 2 illustrates the 95th-percentile amplitude of strain rate across the fiber length after FK-filtering for particular time windows in both the Svalbard and Monaco–Italy datasets. To ensure that the amplitude of fin whale calls is distinguishable from background noise, we limit the analysis to the 5–115 km section of the cable. Noise levels outside this range are too high to detect fin whale calls reliably.

For batch processing, the data is divided into 30-s chunks with a 10-s overlap between consecutive chunks. The overlap ensures that calls occurring at chunk edges are not missed. This setup can result in the same call appearing in two adjacent chunks, which is handled later in the processing pipeline (Sec. V). A bounding box is drawn when a coherent hyperbolic or V-shaped arrival is visible above the local background in the FK-filtered t–x panel. This manual annotation is carried out using the open-source tool Computer vision annotation tool (CVAT Open Source Community, 2025). Although we aim to capture as many vocalizations as possible, this process is subjective and may miss faint calls that are not visually obvious. The resulting dataset consists of 1289 chunks containing a total of 5707

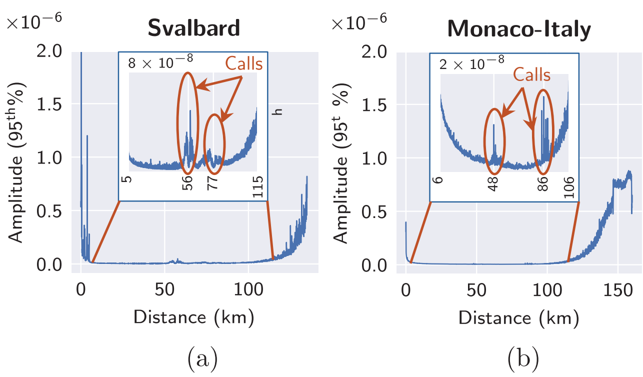

FIG. 2. (a) The 95th-percentile amplitude of strain rate along the cable after FK filtering for the Svalbard dataset from 12:25:39 to 12:26:09 on 22 August 2022. Fin whale calls are observed near 56 and 77 km. Background noise increases near the cable ends, suppressing calls before 5 km and beyond 115 km. (b) Same as (a) but for the Monaco–Italy dataset, 95th-percentile amplitude from 06:10:01 to 06:11:01 on 29 January 2025. Fin whale calls are observed near 48 and 86 km, with background noise similarly masking calls before 6 and beyond 106 km.

vocalizations. The first seven hours, comprising 3588 calls, are used for training. The next three hours, containing 1317 calls, are allocated for validation, while the final two hours, with 802 calls, are reserved for testing.

C. Monaco–Italy data

We additionally use DAS data from the de-commissioned submarine telecommunication cable connecting Monte Carlo, Monaco to Savona, Italy [Figs. 1(b) and 1(d)]. The cable is 162 km long and has been operational since January 1995. It is owned and operated by Sparkle and Monaco Telecom (TeleGeography, 2025). An ASN OptoDAS interrogator was deployed in September 2024. For this study, we use 3 h of data recorded on 29 January 2025, covering a 100 km section of the cable, from 6 to 106 km. This range ensures that the amplitude of fin whale calls is distinguishable from background noise [Fig. 2(b)].

The measurements were acquired at a 312.5 Hz temporal sampling rate. Signals were sampled every 20.43 m over a 40.85 m gauge length. These settings differ from those used in Svalbard. Still, the 312.5 Hz sampling rate is well above the fin whale vocalization band of interest (15–25 Hz), so the calls are sampled without issue. The 20.43 m spatial spacing is also sufficient because call energy can be seen over many kilometers from the closest point on the cable, so features smaller than a meter are not essential. The longer 40.85 m gauge length increases averaging and typically improves SNR. However, the overall SNR can be largely affected by other factors such as patch cables and connectors, fiber coupling, bathymetry, and noise sources.

Each 10 s hierarchical data format version 5 (HDF5) file from Monaco–Italy is smaller (≈ 50 MB vs ≈ 420 MB for Svalbard) due to the sparser sampling. This allows more files to be loaded at once without exhausting memory on the analysis machine (MacBook Pro 2023, Apple, Cupertino, CA; see Sec. V). For testing, we process the Monaco–Italy data in 60 s chunks with a 20 s overlap, different from the 30 s setting used for training, to show that the pipeline works with various chunk lengths. Manual annotations identified 1101 vocalizations across 257 chunks. This dataset is used as an external test set to evaluate the generalization of the model trained on Svalbard data.

III. METHODOLOGY

In this study, we detect fin whale calls in the time–space (t–x) domain to determine each call's arrival time and its position along the cable. We define t as the time axis and x as the fiber–cable axis. Assuming the cable is straight, the arrival pattern of a whale call along the fiber follows a hyperbolic shape

$$\frac{(t-t_0)^2}{\left(\frac{d}{c_s}\right)^2} - \frac{(x-x_0)^2}{d^2} = 1, \tag{3}$$

where x_0 is the cable position closest to the whale, t_0 is the call emission time, d is the distance between the whale and

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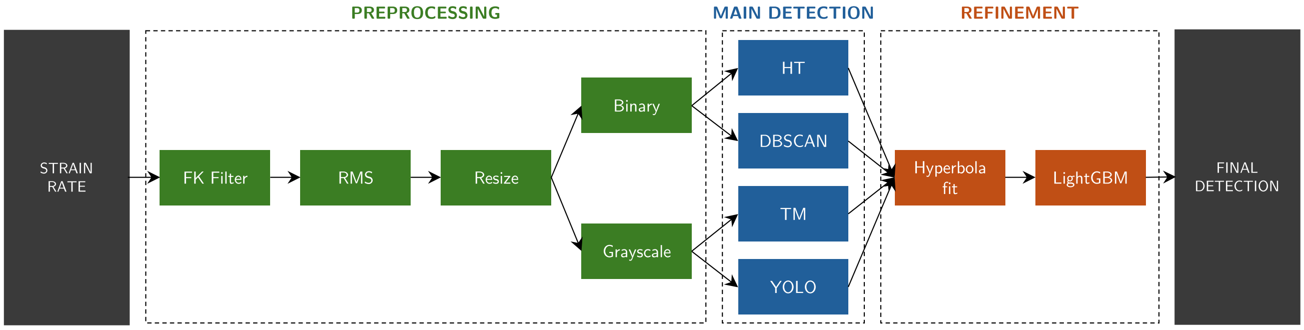

FIG. 3. Workflow for detecting fin whale calls in DAS data. The workflow consists of three main stages: preprocessing, main detection, and detection refinement. In the preprocessing stage, we apply an FK filter to enhance the SNR, compute the RMS amplitude, resize the data, and apply a binary or grayscale transformation. In the main detection stage, we use methods such as HT, DBSCAN, TM combined with DBSCAN, and YOLO to identify fin whale calls. In the detection refinement stage, we fit hyperbolas to the detected calls and calculate residual errors. These errors, along with other features, are used in a LightGBM classifier to further improve the results.

x_0, and c_s is the speed of sound in seawater. The shape of the hyperbola depends on these parameters. For example, a smaller d results in a steeper hyperbola, resulting in a more V-shaped arrival pattern.

The detection workflow has three stages: preprocessing, main detection, and detection refinement (Fig. 3). In preprocessing, we apply an FK filter to enhance the SNR, then compute the root-mean-square (RMS) amplitude, resize the data, and apply binary or grayscale transformation. This produces a compact intensity image with reduced size and complexity for subsequent processing. Because fin whale downsweeps are highly stereotyped and of high contrast in the (t–x) plane, intensity patterns alone are sufficient for detection (Waagaard *et al.*, 2021; Turov *et al.*, 2023). We therefore avoid phase unwrapping—a slow processing step used in phase-based DAS systems—to keep the algorithm fast enough for terabyte-scale datasets. In the main detection stage, we use methods such as HT, DBSCAN, TM followed by DBSCAN, and YOLO to identify fin whale calls. In the detection refinement, we fit hyperbolas to the detected calls and calculate residual errors. These errors, along with other features, are used in a LightGBM classifier to further improve the results.

A. Preprocessing

The preprocessing stage is crucial for enhancing the SNR and preparing the data for effective detection. We apply a series of steps, including FK filtering, RMS filtering, resizing, and foreground transformation.

1. FK filtering

We apply a combined filter in the FK domain, limiting both the frequency band and the along-fiber speed range. The FK domain is obtained by a two-dimensional (2D) Fourier transform of the $t-x$ data; its axes are temporal frequency f (Hz) and along-fiber spatial frequency (wavenumber). We adopt the normalized wavenumber κ (m^{-1}) rather than the angular form. For a sound wave traveling at speed c_s and arriving at the cable with a grazing angle θ, the along-fiber component is

$$\kappa_x = \frac{f}{c_s}\cos\theta(x) = \frac{f}{c_s}\frac{|x-x_0|}{\sqrt{(x-x_0)^2+d^2}}. \quad (4)$$

Because DAS measures axial strain along the fiber, the relevant speed in the FK filter is the apparent along-fiber arrival speed

$$v = \frac{|f|}{|\kappa_x|} = \frac{c_s\sqrt{(x-x_0)^2+d^2}}{|x-x_0|}. \quad (5)$$

Let $\chi = |x-x_0|$. Then,

$$v(\chi, d, c_s) = \frac{c_s\sqrt{\chi^2+d^2}}{\chi}. \quad (6)$$

Fin whales produce low-frequency vocalizations (15–25 Hz) that include downsweeps (25–15 Hz), centered near 20 Hz (Watkins, 1981; Papale *et al.*, 2023). Backbeats (18–20 Hz) also occur, likely for social purposes, while upsweeps can reach 130 Hz during reproductive periods (Papale *et al.*, 2023). Some studies report downsweeps peaking at 45 ± 15 Hz, potentially from overlapping baleen whale signals (Landrø *et al.*, 2022). We therefore use 15–25 Hz as the primary passband and retain coefficients satisfying

$$v_{\min} \le v \le v_{\max}. \quad (7)$$

For reference, the sound speed c_s in seawater depends on salinity, temperature, and pressure (Mackenzie, 1981; Coppens, 1981; Wong and Zhu, 1995). On 22 August 2022, the local temperature was ≈ 6.5 °C (AWIPEV, 2025), salinity ≈ 34.5 PSU, and pressure ≈ 505 kPa at 50 m depth (Korhonen *et al.*, 2024; De Rovere *et al.*, 2022; Bloshkina *et al.*, 2021), yielding $c_s \approx 1{,}477$ m/s (National Physical Laboratory, 2025). With c_s treated as fixed, varies with χ and d according to Eq. (6), as summarized in Table I.

TABLE I. Apparent speed (m/s) as a function of along-fiber offset χ (km) and whale-cable distance d (km).

$\chi\backslash d$	2	5	10	20	50	100
2	2089	3977	7531	14 844	36 955	73 865
5	1591	2089	3303	6090	14 844	29 577
10	1506	1651	2089	3303	7531	14 844
20	1484	1522	1651	2089	3977	7531
50	1478	1484	1506	1591	2089	3303
100	1477	1479	1484	1506	1651	2089

Fin whale detection ranges depend on water depth and environment. Hydrophone studies report detection distances of around 30 km in shallow water and up to 55 km in deep water (Ahonen *et al.*, 2021), while ocean-bottom seismometers have recorded calls at distances of ~ 15 km in deep environments (McDonald and Fox, 1999; Wilcock, 2012). Recent DAS measurements suggest a range up to ~ 40 km (Wilcock *et al.*, 2023). To set bounds for the speed constraint, we take $d_{\max} = 50$ km and $\chi_{\min} = 5$ km, giving $v_{\max} = 14{,}844$ m/s; and $d_{\min} = 5$ km with $\chi_{\max} = 50$ km, giving $v_{\min} = 1{,}484$ m/s (Table I).

Accordingly, we apply an FK filter with a 15–25 Hz band and an apparent speed range of 1484–14 844 m/s. This improves the SNR of fin whale vocalizations while suppressing unwanted signals (Fig. 4), providing cleaner input for subsequent detection steps.

2. RMS filtering

We compute the RMS amplitude using a causal, backward-looking window of N_{rms} samples. Here, $N_{\mathrm{rms}} = f_s \times 0.5$, corresponding to 0.5 s at sampling rate f_s, which is 625 Hz for Svalbard and 312.5 Hz for Monaco–Italy. A fin whale downsweep in our data is typically about 1 s long, so the RMS window must be shorter to avoid over-smoothing but still long enough to reduce noise. We chose 0.5 s as a compromise after empirical testing. This step boosts SNR and highlights fin whale calls. For each position along the cable at time t_i, the RMS amplitude $M_{\mathrm{rms}}(t_i)$ is calculated as

$$M_{\mathrm{rms}}(t_i) = \sqrt{\frac{1}{N_{\mathrm{rms}}} \sum_{k=i-N+1}^{i} M^2(t_k)}, \tag{8}$$

where $M(t_k)$ is the amplitude at time t_k. This causal window ensures that the RMS amplitude is computed using past and present data, avoiding future information leakage.

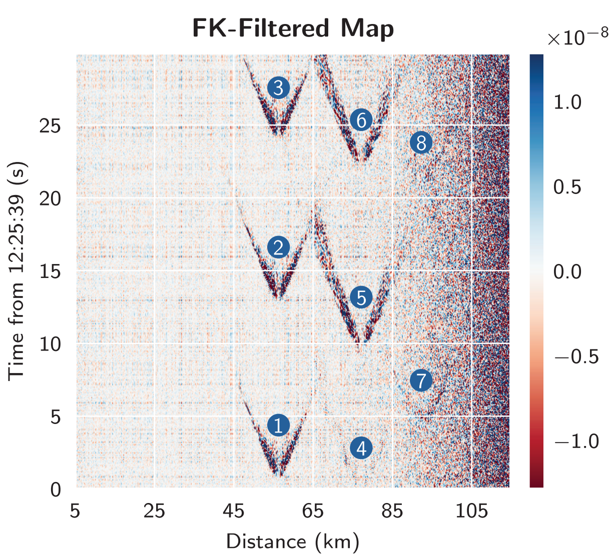

FIG. 4. The t–x domain after FK filtering (15–25 Hz, 1484–14 844 m/s) in the example windows in Svalbard. The x-axis represents distance along the cable from the interrogator, while the t-axis corresponds to time on 22 August 2022. At least eight fin whale calls are visible: three near 56 km (calls 1–3), three near 77 km (calls 4–6), and two near 95 km (calls 7 and 8). The three calls at 56 km and the last two at 77 km are clear, while the remaining three are less noticeable.

3. Resize and foreground transformation

We resize the data to 640×640 pixels using linear interpolation to standardize the input for all detection methods. Next, we apply either a binary or grayscale transformation, depending on the method: binary for DBSCAN and HT, and grayscale for TM and YOLO. These are performed in a column-wise manner, treating each spatial point along the cable independently to enhance robustness against location-specific noise. For the binary transformation, we retain the top 5% of values in each column, setting them to 1 and the rest to 0. The percentile is computed for each 30-s window, so the threshold adapts to time and sensing location. For the grayscale transformation, each column is normalized to the range 0–255.

B. Main detection

After separating the foreground from the background using binary or grayscale processing, we apply four different detection methods: HT (Duda and Hart, 1972), DBSCAN (Ester *et al.*, 1996), TM (Lewis, 1995), and YOLO (Redmon *et al.*, 2016). These methods differ in how they detect foreground features: HT and TM are classical image processing techniques, with HT employing line-based voting and TM using pattern matching; DBSCAN is an unsupervised machine learning method based on density clustering; and YOLO is a deep learning-based object detection algorithm. Because YOLO leverages deep learning and training data, it may have an advantage in performing detection compared to the other methods. However, we include all four approaches to provide a broader comparison and to highlight the relative strengths and limitations of each technique in detecting fin whale calls in the DAS data.

1. Hough transform

As shown in Fig. 4, the hyperbolic arrival pattern of fin whale calls often appears as two nearly straight line segments forming a V-shape. This occurs because, beyond approximately 5 km from the cable, the grazing angle of the sound waves changes minimally, especially for whales close to the cable. Consequently, the hyperbolic curve flattens and can be well-approximated by straight-line segments.

The HT is a robust technique for detecting parameterized shapes in images (Duda and Hart, 1972), particularly those defined by a small number of parameters, such as lines and circles. In its general form, the shape to be detected is described by an implicit equation

$$F(x, t, a_1, a_2, \ldots, a_n) = 0, \tag{9}$$

where (x, t) are image coordinates, and $\{a_1, a_2, \ldots, a_n\}$ are the parameters describing the shape. The method begins by constructing a binary foreground mask that highlights relevant pixels, such that the set of all foreground points is

$$\{(x, t) \mid \text{pixel at } (x, t) \text{ is an edge or foreground point}\}.$$

Each parameter a_j is discretized into a finite set of possible values $\left\{ a_j^{j_1}, a_j^{j_2}, \ldots, a_j^{j_{K_j}} \right\}$. This discretization defines an n-dimensional accumulator space, one axis per parameter, where votes are accumulated.

To map the image space to the parameter space, one rearranges Eq. (9) to express a chosen parameter a_i as a function of the remaining parameters and the pixel coordinates

$$a_i = G\big(x, t, \{a_j | j \neq i\}\big). \tag{10}$$

For each foreground pixel (x, t), one iterates over the discrete values of all parameters $\{a_j\}$ except a_i, substitutes these values into Eq. (10) to compute $\hat{a}_i = G(x, t, \{a_j = a_j^{j_k} \mid j \neq i\})$, rounds $\hat{a}_i$ to the nearest discrete bin $a_i^{i_k}$, and increments the corresponding cell of the accumulator,

$$\begin{aligned} &A(a_1^{1_k}, a_2^{2_k}, \ldots, a_i^{i_k}, \ldots, a_n^{n_k}) \\ &\quad \leftarrow A(a_1^{1_k}, a_2^{2_k}, \ldots, a_i^{i_k}, \ldots, a_n^{n_k}) + 1. \end{aligned} \tag{11}$$

After every foreground pixel has voted, a threshold $V_{\min}$ is applied to filter out parameter combinations with low vote counts,

$$A(a_1, a_2, \ldots, a_n) > V_{\min}, \tag{12}$$

and non-maximum suppression is performed to isolate prominent peaks and avoid multiple detections arising from minor variations in the parameters.

The basic algorithm can be computationally expensive, with complexity $\mathcal{O}(M \prod_{j \neq i} K_j)$ where M foreground pixels and K_j is the number of bins for the parameter a_j. This can be mitigated using gradient-based constraints, hierarchical search methods, and adaptive binning. These refinements help reduce the dimensionality of the accumulator space and mitigate quantization errors, enabling the HT to remain a robust and versatile tool for shape detection in a wide range of imaging applications.

The standard Cartesian form of a line is $t = mx + b$, with parameters (m, b). However, the polar form, $r = x \cos\theta + t \sin\theta$, with parameters (r, θ), is more commonly used as it avoids issues with vertical lines and provides a more stable representation for voting in the accumulator space (Duda and Hart, 1972).

We use a probabilistic version of the HT (Matas *et al.*, 2000), which is more robust to noise and outliers. The algorithm works by randomly sampling a subset of points from the input data and voting in the accumulator space. The distance resolution is set to $\rho = 1$ pixel, and the angular resolution is $\theta = 0.02$ radians. The minimum line length is set to 4000 m, with $V_{\min}$ set to be equivalent to 75% of the minimum line length. Additionally, `maxLineGap`, which defines the maximum gap between segments to merge, is set to 20% of the line length. These values were determined empirically by testing a range of configurations and selecting those that provided reliable detection of fin whale calls while minimizing false positives across time windows and study sites.

2. DBSCAN

The objective is to cluster spatially proximate foreground points in a foreground mask, where each cluster is expected to represent a vocalization event. Several clustering methods, including K-Means, Gaussian mixture model (GMM), DBSCAN, hierarchical density-based clustering (HDBSCAN), and ordering points to identify clustering structure (OPTICS), can be applied to this task. Since the number of vocalizations is unknown, clustering methods that require a predefined number of clusters, such as K-Means and GMM, are not suitable. HDBSCAN and OPTICS are designed for datasets with varying density clusters (McInnes *et al.*, 2017; Ankerst *et al.*, 1999), whereas foreground masks typically have uniform event density. Therefore, we employ an adaptive DBSCAN for spatiotemporal clustering.

DBSCAN is a density-based clustering algorithm that identifies clusters based on dense regions in the data (Ester *et al.*, 1996). It has been widely used for event detection in geoscience applications (Hafezi *et al.*, 2019; Stinco *et al.*, 2021). The algorithm requires two parameters: ϵ, the radius of the neighborhood, and *MinPts*, the minimum number of points required to form a dense region.

Formally, let $\mathcal{P} = \{p_1, p_2, \ldots, p_n\}$ be the set of all foreground points, where each $p_i = (x_i, t_i) \in \mathbb{R}^2$. The ϵ-neighborhood of a point p_i is defined as

$$N_\epsilon(p_i) = \{p_j \in \mathcal{P} | \delta(p_i, p_j) \leq \epsilon\}, \tag{13}$$

where $\delta(p_i, p_j)$ is a distance metric between points p_i and p_j.

Based on this neighborhood definition, DBSCAN classifies points into three categories. A core point is any point that has at least *MinPts* neighbors within distance ϵ, including itself

$$|N_\epsilon(p_i)| \geq MinPts. \tag{14}$$

A border point is a point that lies within the ϵ-neighborhood of at least one core point but does not satisfy the core point condition, meaning it has fewer than *MinPts* neighbors,

$$\exists p_i \in \mathcal{P} \text{ such that } p_j \in N_\epsilon(p_i) \text{ and } |N_\epsilon(p_j)| < MinPts. \quad (15)$$

Finally, any point that is neither a core point nor a border point is classified as noise. Core points that are within each other's ϵ-neighborhood or are connected through other core points (i.e., density-reachable) belong to the same cluster. Each border point is assigned to the cluster of at least one core point within its ϵ-neighborhood. Noise points remain unassigned and do not belong to any cluster. Figure 5 illustrates core, border, and noise points in DBSCAN.

The clustering parameters were chosen empirically with careful consideration to balance detection performance across different time windows. After extensive testing, we set $\epsilon = 26$ and $MinPts = 100$, aiming to maximize true detections of fin whale calls. This also leads to a substantial number of false positives, typically appearing as small clusters, which are effectively filtered out later in the detection refinement stage (Sec. III B). Because DBSCAN works in pixel units, the prior RMS and resize steps set the pixel scale and point density, which directly influence the choice of ϵ and $MinPts$. Future work could explore adaptive parameter selection based on data characteristics or automatic tuning (e.g., tree-structured Parzen estimator). In addition, a key aspect is defining an appropriate distance metric δ that balances the temporal and spatial axes, ensuring that points lying along a path with a given apparent speed v are considered close. To achieve this balance, we rescale the spatial coordinate by

$$\alpha = \frac{\Delta x}{\Delta t\, v},$$

where Δx is the spatial sampling interval and Δt is the temporal sampling interval, and we set $v = 1,750\,\mathrm{m/s}$. Then, for two points $p_i = (x_i, t_i)$ and $p_j = (x_j, t_j)$, we define the rescaled Euclidean distance as

$$\delta(p_i, p_j) = \sqrt{\left[\alpha\,(x_i - x_j)\right]^2 + (t_i - t_j)^2}. \quad (16)$$

The choice of is not highly sensitive—offsets of a few hundred meters per second have minimal impact because the binary foreground mask already filters out most noise, and the DBSCAN algorithm is robust to small variations in the distance metric.

3. Template matching

TM and its variants have been successfully employed to detect blue whale and minke whale vocalizations in both the time domain and the time–frequency domain (Bouffaut *et al.*, 2018; Ogundile and Versfeld, 2020).

In this study, we apply TM directly in the $t - x$ domain. TM is a technique from image processing that locates a small template within a larger search image by identifying the position (x, t) where the template best aligns with the image.

The template $T(x', t')$ is designed to mimic the hyperbolic shape of fin whale calls, with an apparent speed of $v = 1,750$ m/s. The template height is set to 4 s, and the signal thickness is 0.5 s (Fig. 6).

The formulation of TM is defined as follows. Let $I(x, t)$ represent the intensity of the search image at pixel (x, t), and $T(x', t')$ denote the intensity of the template at pixel (x', t'). The dimensions of the template are $w \times h$. The mean intensity of the template is given by $\overline{T} = (1/w \cdot h) \times \sum_{x'=0}^{w-1} \sum_{t'=0}^{h-1} T(x', t')$. Similarly, the mean intensity of the $w \times h$ patch of the image starting at (x, t) is computed as $\overline{I}_{(x,t)} = (1/w \cdot h) \sum_{x'=0}^{w-1} \sum_{t'=0}^{h-1} \times I(x + x', t + t')$. The matching score at location (x, t) is represented by $R(x, t)$.

Several methods exist for computing the score $R(x, t)$, including the sum of squared differences, sum of absolute differences, cross correlation, normalized cross correlation, and zero-mean normalized cross correlation (ZNCC). In this study, we focus on ZNCC, which is more robust to local brightness variations (Lewis, 1995). Its formulation is

$$R(x, t) = \frac{\sum_{x',t'} [T'(x', t') \cdot I'(x + x', t + t')]}{\sqrt{\left[\sum_{x',t'} T'(x', t')^2\right]\left[\sum_{x',t'} I'(x + x', t + t')^2\right]}}, \quad (17)$$

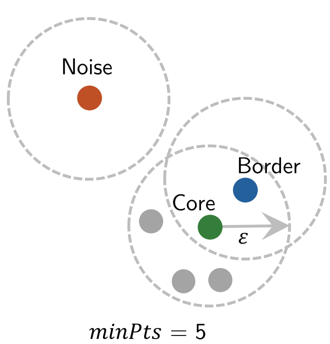

FIG. 5. Illustration of core, border, and noise points in DBSCAN. The green point represents a core point with $MinPts = 5$. The blue point is a border point, while the orange point is classified as noise.

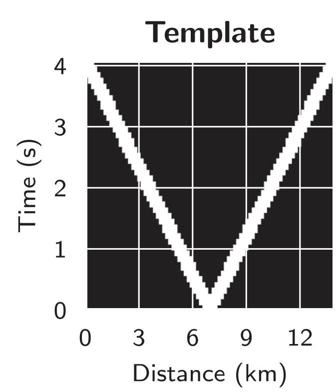

FIG. 6. Template used for TM in the t–x domain. The template mimics the hyperbolic shape of fin whale calls, with an apparent speed of $v = 1750$ m/s. The template height is 4 s, and the signal thickness is 0.5 s.

where

$$T'(x',t') = T(x',t') - \overline{T}, \tag{18}$$

$$I'(x+x',\, t+t') = I(x+x',\, t+t') - \overline{I}_{(x,t)}. \tag{19}$$

By subtracting the mean from both the template and the corresponding image patch, we zero-center the data. Dividing by their magnitudes normalizes the result, yielding a correlation coefficient in the range $[-1, 1]$. A value near 1 indicates a strong match, 0 indicates no correlation, and -1 indicates inverse correlation.

After computing the matching score $R(x,t)$ for all valid (x,t) positions in the image, we retain the set of pixels where the score exceeds a predefined threshold $\tau = 0.2$, defined as

$$S = \{(x,t) | R(x,t) \geq \tau\}. \tag{20}$$

We finally apply DBSCAN to the set S to cluster the detected points. The clustering process groups nearby points, which likely belong to the same vocalization event. Each cluster is then treated as a potential fin whale call. For clarity, throughout the remainder of this paper, we use TM to refer to the full process of template matching followed by DBSCAN clustering.

4. YOLO

A final method we use is a convolutional neural network (CNN)-based approach. Advances in deep learning have significantly improved detection in underwater acoustic environments. Examples include a detection transformer for beluga whale call detection in hydrophone data (Cotillard *et al.*, 2024), YOLO for whale detection in satellite imagery (Kapoor *et al.*, 2023; Green *et al.*, 2023), and long short-term memory architectures for dolphin detection in UAV videos (Canelas *et al.*, 2025). We choose YOLO for its balance between speed and accuracy (Vijayakumar and Vairavasundaram, 2024), which is essential for real-time deployment in the interrogator.

YOLO is a real-time object detection algorithm that formulates detection as a single-stage regression problem. Unlike traditional methods, which separate object classification and localization, YOLO predicts bounding boxes and class probabilities in a single forward pass, enabling high-speed detection. Introduced by Redmon *et al.* (2016), the original YOLO model divided the input image into an $S \times S$ grid, with each cell predicting B bounding boxes and C class probabilities. The final output is a tensor of dimensions $S \times S \times [B \times (5 + C)]$, where the five parameters represent bounding box coordinates, width, height, and confidence score. The model is trained using a multi-component loss function that balances localization accuracy, confidence prediction, and classification performance.

Compared to TM, which performs a 2D convolution with a fixed kernel, YOLO uses multiple learnable convolutional kernels. These kernels are optimized end-to-end during training, so the network learns features that respond to fin whale arrival patterns across scales, SNRs, and backgrounds. TM may work well when the target pattern is consistent, while YOLO is expected to be more robust to variability.

We employ YOLOv11, released by Ultralytics in September 2024, which enhances accuracy and efficiency by improving localization and classification while reducing computational cost. It achieves a higher mean average precision (mAP) on benchmark datasets with fewer parameters than previous versions and expands to segmentation, pose estimation, and oriented detection, increasing its versatility (Jocher and Qiu, 2024).

Given the GPU's memory limitation of 12 GB, we utilize the YOLOv11 large model, which has 25.3×10^6 parameters and requires approximately 10 GB of memory during training. We set the batch size to 16 and train for 100 epochs using the AdamW optimizer. To enhance generalization, the training process incorporates mosaic augmentation.

C. Detection refinement

The detection methods described previously produce bounding boxes for candidate fin whale calls, but these may include false positives that do not follow the expected hyperbolic pattern in the *t*–*x* domain. To address this, we fit a hyperbola to the foreground points within each detected bounding box and calculate residual errors. These errors, along with additional features from the detection stage, are used as input to a LightGBM classifier.

1. Hyperbola fitting

The fitting process involves estimating the parameters of the hyperbola defined in Eq. (3) using a least squares approach. The goal is to minimize the error between the observed data points and the predicted hyperbola.

For each detected bounding box, we extract the set of foreground points (x_i, t_i) and fit the hyperbola model in Eq. (3) using least squares. The quality of the fit is evaluated using the normalized root mean square error (RMSE) and normalized mean absolute error (MAE), where normalization is performed with respect to the bounding box height h_{box},

$$\text{n} - \text{RMSE} = \frac{1}{h_{\text{box}}} \sqrt{\frac{1}{n_{\text{f}}} \sum_{i=1}^{n_{\text{f}}} \left[t_i - \hat{t}(x_i)\right]^2}, \tag{21}$$

$$\text{n} - \text{MAE} = \frac{1}{h_{\text{box}}} \frac{1}{n_{\text{f}}} \sum_{i=1}^{n_{\text{f}}} |t_i - \hat{t}(x_i)|, \tag{22}$$

where n_{f} is the number of foreground points in the bounding box, and $\hat{t}(x_i)$ is the predicted value from the fitted hyperbola model [Eq. (3)]. The fitted parameters are x_0, t_0, d, and c_s. The normalized RMSE is more sensitive to outliers, while the normalized MAE reflects the average fitting error. Both metrics are used to assess the quality of the hyperbola fit in this study. Figure 7 shows two examples of hyperbola fitting results for detected calls 5 and 7 from Fig. 4. Call 5 is a clear call with a good fit, indicated by low normalized

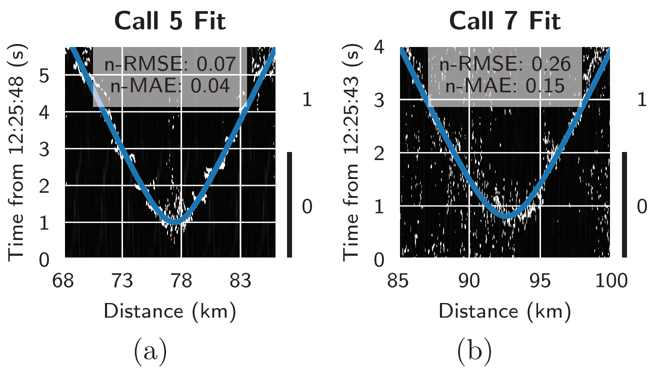

FIG. 7. Hyperbola fitting results for two example calls. The blue curves show the fitted hyperbolas overlaid on the foreground points within the detected bounding boxes. (a) Call 5 demonstrates a strong, well-defined detection with a good fit, indicated by low normalized RMSE and MAE. (b) Call 7 represents a weaker, less distinct detection with a poorer fit, reflected in higher normalized RMSE and MAE.

RMSE and MAE, while call 7 is more vague, reflected in higher normalized RMSE and MAE.

2. LightGBM

After applying the detection methods and fitting hyperbolas, we obtain several features for each detected bounding box. These features include the bounding box coordinates, width, height, and the normalized RMSE, MAE from the hyperbola fitting process, and other features from the detection depending on the methods. The set of features is then used to train a LightGBM classifier to distinguish between true positive (TP) and false positive (FP) detections.

A detected bounding box from the main detection stage is considered a TP if its intersection over union (IoU) with a ground truth box is greater than or equal to 0.25, and a FP otherwise. The IoU is defined by

$$\mathrm{IoU} = \frac{\text{Area of intersection}}{\text{Area of union}}. \tag{23}$$

Here, the threshold is lower than the standard 0.5 used in general object detection tasks to account for the subjectivity in defining bounding boxes for hyperbolic shapes. The classifier is trained on the first seven hours of Svalbard data and validated on the following three hours as described in Sec. III B. We optimize the classifier based on the F1 score, which balances precision and recall

$$\text{precision} = \mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \tag{24}$$

$$\text{recall} = \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \tag{25}$$

$$F1 = 2\cdot\frac{P\cdot R}{P+R}, \tag{26}$$

where false negative (FN) refers to instances where a true call is not detected during the main detection stage.

LightGBM is a gradient boosting framework that builds decision trees in a leaf-wise manner. It is efficient and well-suited for large datasets, using histogram-based optimization and feature bundling to reduce memory and training time (Ke *et al.*, 2017).

IV. RESULTS

A. Hough transform

Figure 8 illustrates the detection results in the same time windows as shown in Fig. 4. For calls 1, 2, 3, 5, 6, and 7, HT successfully detects both sides of the hyperbola, while it misses call 4. The missed detection occurs because call 4 is relatively weak compared to other calls. It is close to the background noise, resulting in insufficient pixel density after the binary transformation for HT to form a line. For call 8, it detects only one side.

A notable issue is that HT often detects multiple lines for the same side of a call [Fig. 8(b)]. To address this, we group overlapping lines and retain only one. Let L and L' be two line segments, defined by their starting and ending points (x_1, t_1), (x_2, t_2) and (x'_1, t'_1), (x'_2, t'_2), respectively. The distance D between the two lines is defined as the average vertical distance between them over the overlapping interval,

$$D(L, L') = \frac{1}{x_{\max} - x_{\min}} \int_{x_{\min}}^{x_{\max}} |L(x) - L'(x)|\, dx, \tag{27}$$

where $x_{\min} = \max(x_1, x'_1)$ and $x_{\max} = \min(x_2, x'_2)$. If $x_{\min} > x_{\max}$, the segments do not overlap and we set $D = \infty$.

To merge nearby line segments, we apply single-linkage clustering using this distance metric with a threshold of one second [Fig. 8(c)]. After clustering, we pair lines with close starting points (x_1, t_1) to form the two sides of a hyperbolic arrival, representing a single whale call [Fig. 8(d)].

Table II summarizes the detection performance on the Svalbard test set. The HT line detection method achieves a precision of 0.493, recall of 0.499, and an F1 score of 0.496, indicating it successfully detects approximately half of the true calls while producing a similar number of FPs. Depending on the application, the parameters of the HT algorithm can be adjusted to prioritize either precision or recall. Additionally, we compute the mean F1 score across different IoU thresholds, ranging from 0.25 to 0.75 with a step of 0.05, to provide a more robust metric against IoU sensitivity. The resulting mean F1 score is 0.247.

The post-refinement step significantly improves the precision to 0.911, while the recall drops slightly to 0.416, resulting in an F1 score of 0.571 and a mean F1 score of 0.322.

B. DBSCAN

As shown in Fig. 9, DBSCAN successfully detects seven out of eight fin whale calls, missing only call 4. Similar to the HT case, this call is weak and close to background noise, leaving almost no pixels after the binary transformation for DBSCAN to form a cluster. In addition, calls 2 and 5, as well as calls 3 and 6, are grouped together into single clusters. This occurs because the right side of the hyperbola for calls 2 and 3 overlaps with the left side of the hyperbola for calls 5 and 6,

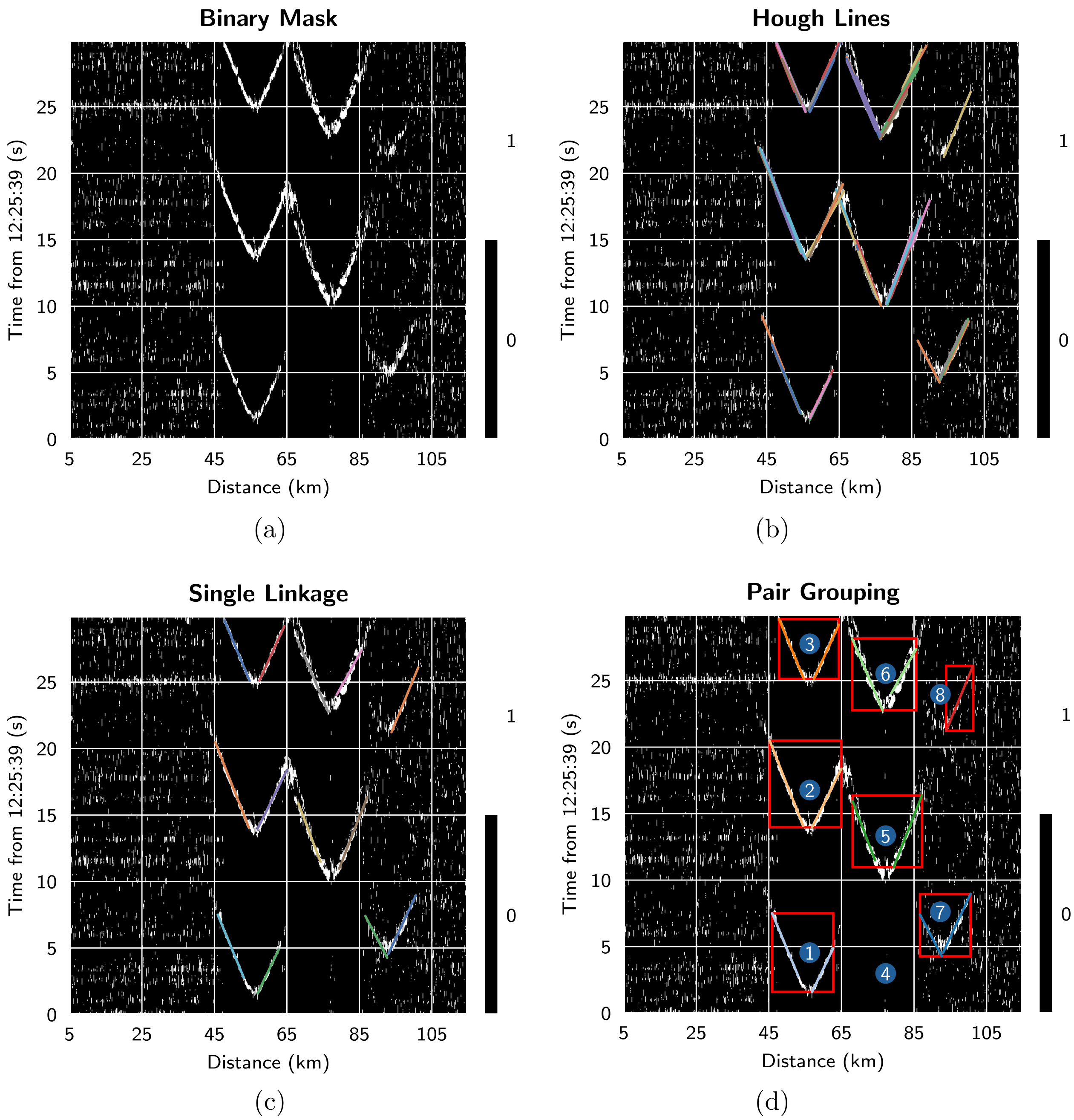

FIG. 8. HT line detection process. (a) Binary foreground image obtained after preprocessing. (b) HT detection results overlaid on the binary foreground image. (c) Detected lines grouped using single-linkage clustering to remove duplicates. (d) Paired lines forming V-shaped patterns representing fin whale calls. If pairing is not possible (e.g., call 8), the lines are retained as individual detections for further refinement.

TABLE II. Detection performance metrics for all methods on the Svalbard test set. *Main* refers to results from the initial detection stage, and *Post* refers to results after the post-refinement step.

Method	Stage	TP	FP	FN	P	R	F1	mF1
HT	Main	640	659	642	0.493	0.499	0.496	0.247
	Post	533	52	749	0.911	0.416	0.571	0.322
DBSCAN	Main	559	7015	723	0.074	0.436	0.126	0.050
	Post	392	144	890	0.731	0.306	0.431	0.192
TM	Main	888	251	394	0.780	0.693	0.734	0.325
	Post	752	84	530	0.900	0.587	0.710	0.326
YOLO	Main	1135	170	147	0.870	0.885	0.877	0.758
	Post	1092	78	190	0.933	0.852	0.891	0.776

respectively. The density at these intersections is sufficient for DBSCAN to link them via density-based connectivity. Consequently, each cluster representing calls 2–5 and 3–6 results in one TP and one FP.

Additionally, DBSCAN introduces many FPs between 5 and 45 km along the cable and beyond 100 km, due to noise in the foreground mask. However, these FPs are relatively small and can be effectively mitigated through post-processing steps.

The detection performance of DBSCAN on the Svalbard test set is presented in Table II. DBSCAN achieves a precision of 0.074, recall of 0.436, and an F1 score of 0.126, with a mean F1 score of 0.050. The low precision is

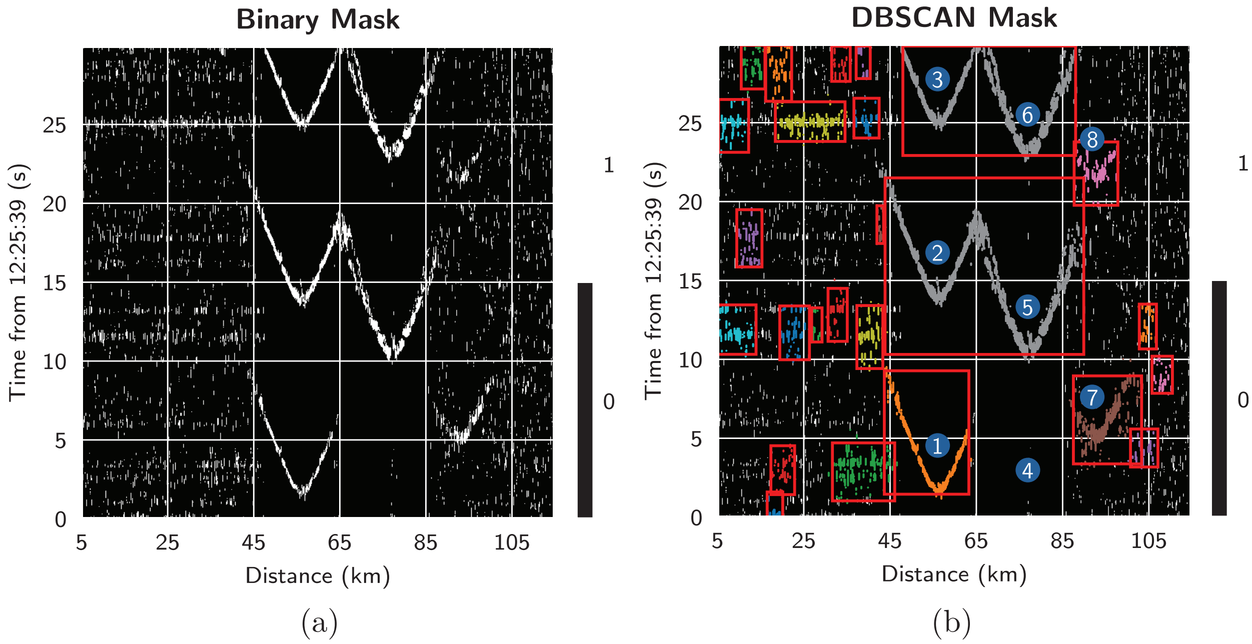

FIG. 9. DBSCAN detection process. (a) Binary foreground image obtained after preprocessing. (b) DBSCAN clustering results overlaid on the binary foreground image. Dense point clusters correspond to fin whale calls, and bounding boxes are drawn around the detected clusters.

primarily attributed to the high number of FPs introduced by the clustering process. However, the post-refinement step significantly enhances the precision to 0.731, while the recall dropped to 0.306. This results in an improved F1 score of 0.431 and a mean F1 score of 0.192.

C. Template matching

As shown in Fig. 10, TM successfully detects seven out of eight fin whale calls, missing only call 4. Importantly, it does not introduce any FPs. Calls 1, 2, 3, 5, and 6 are detected with high confidence, as the binary foreground mask clearly preserves the V-shape of the hyperbola. However, calls 7 and 8 are less distinct, with only a few pixels preserved in the binary foreground mask. Lowering the threshold could preserve more pixels for calls 7 and 8, but this would likely introduce additional FPs.

As summarized in Table II, the TM method achieves a precision of 0.780, a recall of 0.693, and an F1 score of 0.734. The mean F1 score is 0.325, indicating a good balance between precision and recall.

The post-refinement step improves the precision to 0.900, but the recall drops to 0.587, resulting in an F1 score of 0.710 and a mean F1 score of 0.326. This suggests that the post-refinement step is less effective for TM. The TM method already produces high-quality detections with minimal FPs (high precision), and the post-refinement step, fit on the training and validation sets, may not generalize as well to the test set. The post-refinement step is more beneficial for methods that initially produce a higher number of FPs.

D. YOLO

YOLO successfully detects seven out of eight fin whale calls without introducing any FP (Fig. 11). This performance is highly dependent on the quality of the training data, particularly the annotations, and the similarity between the training and test datasets.

Table II shows that YOLO achieves a precision of 0.870, recall of 0.885, and an F1 score of 0.877. The mean F1 score is 0.758. This indicates that YOLO is highly effective for fin whale detection in the $t-x$ domain, achieving a high precision while maintaining good recall. The post-refinement step further improves the precision to 0.933, while the recall drops slightly to 0.852, resulting in an F1 score of 0.891 and a mean F1 score of 0.776. This improvement is rather small because YOLO already produces high-quality detections with minimal FPs. The strong performance of YOLO is expected, as it leverages deep learning to learn complex patterns from the training data, allowing it to generalize well to testing data. It is worth noting that the imperfect precision of 0.933 is not necessarily an issue. Since the training data are manually labeled, some detections counted as FPs can actually be true calls that were missed during annotation, as confirmed through our post-checks.

E. Comparison

Among the evaluated methods, Table II highlights that DBSCAN is the least effective, with a precision of 0.074 due to a high number of FPs, resulting in an F1 score of 0.126. The HT method shows moderate performance, achieving a recall of 0.499 and a precision of 0.493, leading to an F1 score of 0.496. TM demonstrates the best performance among the non-deep learning methods, with a precision of 0.780 and recall of 0.693, yielding an F1 score of 0.734. However, YOLO surpasses all other methods, achieving the highest precision (0.870) and recall (0.885), resulting in an F1 score of 0.877. Notably, while the F1 score of

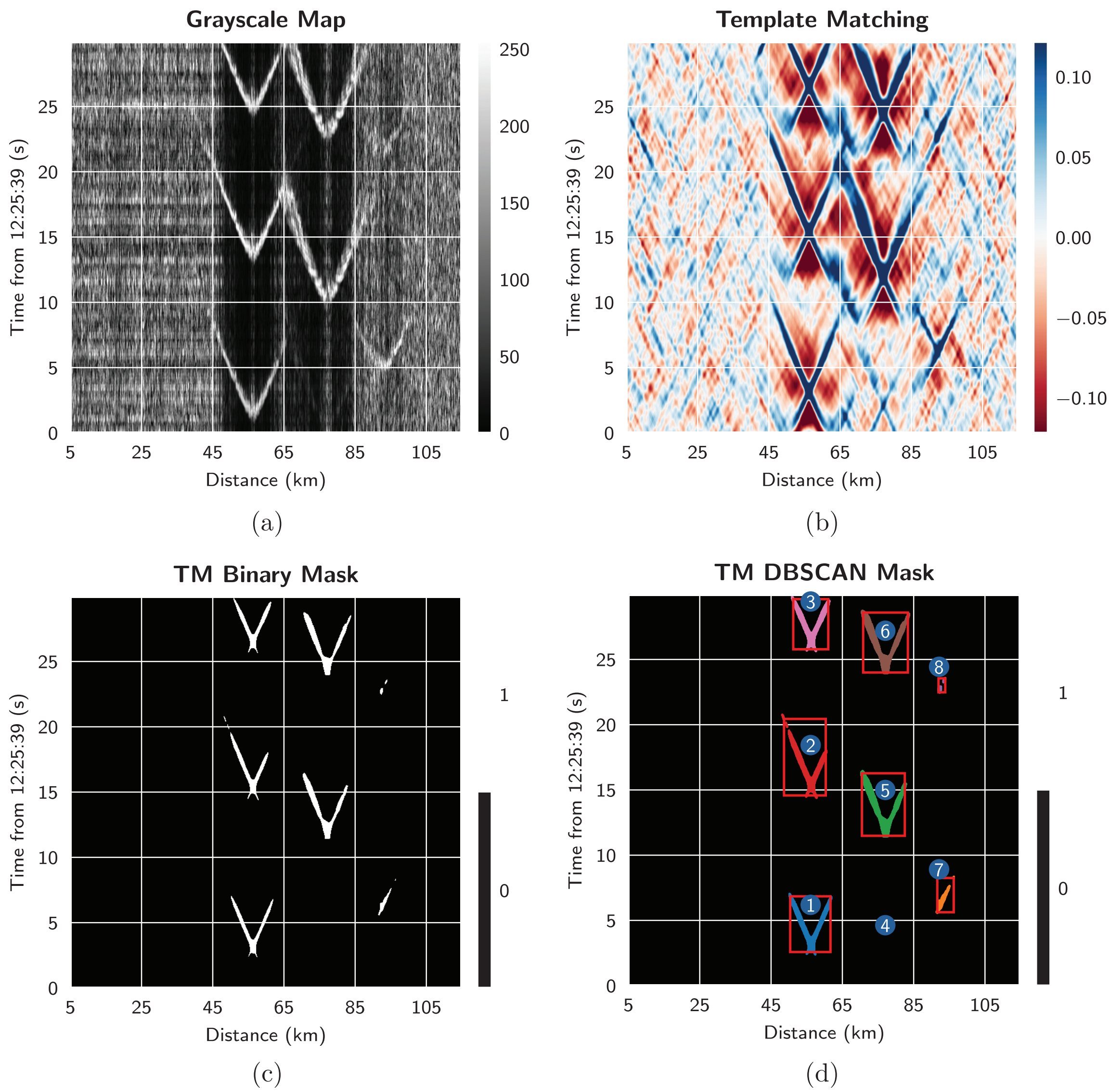

FIG. 10. TM detection process. (a) Grayscale image obtained after preprocessing. (b) TM matching score, obtained by sliding a V-shaped template—representing the expected fin whale call trajectory (velocity $v = 1750$ m/s)—across the grayscale image. Higher intensity indicates stronger template matches. (c) Thresholded TM score map showing only high-response regions retained for further processing. (d) DBSCAN applied to the high-score points, with bounding boxes drawn around the resulting detected clusters.

YOLO is approximately 0.14 higher than that of TM, the mean F1 score difference is substantial, with YOLO achieving 0.758 compared to TM's 0.325. This indicates that YOLO is considerably more robust across varying IoU thresholds.

Table II also presents the detection performance after applying the post-refinement step using LightGBM. The refinement significantly improves the precision of all methods, with YOLO achieving the highest precision of 0.933, followed by HT at 0.911, TM at 0.900, and DBSCAN at 0.731. This improvement aligns with the refinement step's design, which filters out FPs based on hyperbola fitting and other features. However, this comes at the cost of slightly reduced recall, as some TPs may be filtered out. Among the methods, YOLO maintains the highest recall at 0.852, while TM and HT achieve recalls of 0.587 and 0.416, respectively. DBSCAN has the lowest recall at 0.306.

A key preprocessing step involves applying column-wise binary or grayscale transformation to the t–x domain data before detection, rather than using an image-wise approach. This offers two main advantages. First, it ensures robustness against noisy spatial points, such as the initial near-shore section or the far end of the cable, which is critical for deployment in environments with varying noise levels. This approach allows models trained on one location to generalize effectively to detect calls in another DAS system location. Second, the column-wise transformation enhances the contrast of the hyperbolic shapes of fin whale calls, making them more distinguishable from background noise. A comparison of the detection performance of YOLO on Svalbard test data using column-wise versus image-wise transformation is shown in Table III. The column-wise approach consistently outperforms the image-wise method across all metrics, including Precision, Recall, F1, and mF1,

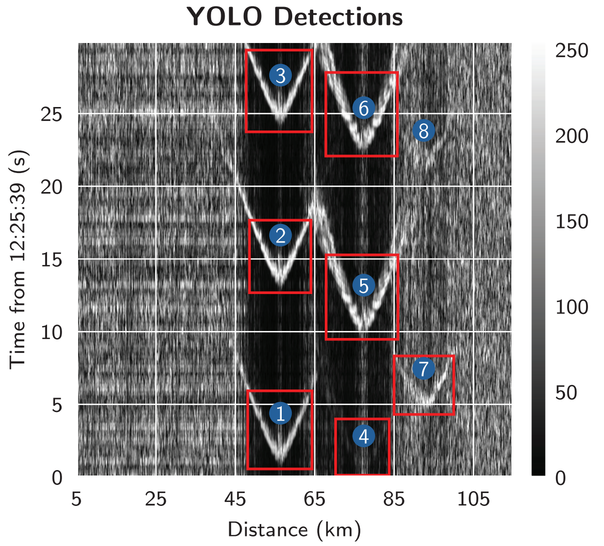

FIG. 11. YOLO detection results. The YOLO model detects seven out of eight fin whale calls and does not introduce any FP. The model is trained for seven hours and can generalize well to the test set.

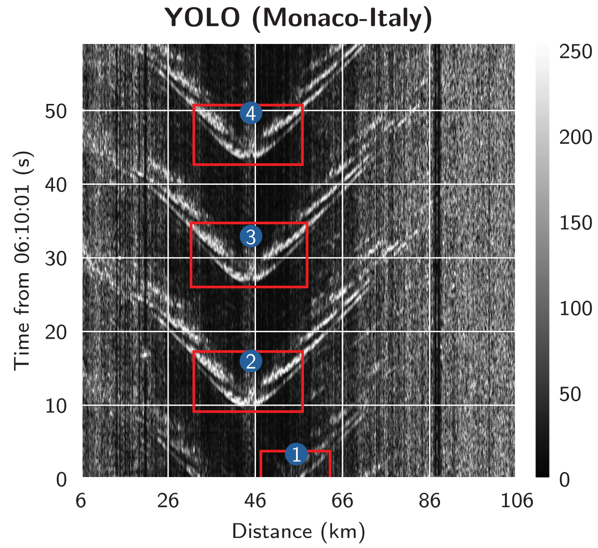

FIG. 12. YOLO detection results on the Monaco–Italy dataset. The model detects three complete fin whale calls, with one additional call partially captured due to limited data in the current batch. The bounding boxes are drawn around the detected calls, demonstrating the model's effectiveness in identifying fin whale calls in this dataset.

demonstrating its effectiveness in improving detection accuracy.

F. Monaco–Italy dataset

Given YOLO's superior performance compared to other methods, it was applied to the Monaco–Italy dataset. An example of the detection results is shown in Fig. 12. The model successfully detects all three complete fin whale calls present in the image (Calls 2, 3, and 4). One additional call is only partially captured due to limited data in the current batch (Call 1), but it would likely be detected in the preceding batch during continuous processing.

Table IV summarizes the performance of YOLO on the Monaco–Italy dataset. Without additional fine-tuning, the model achieved an F1 score of 0.801, demonstrating robust generalization despite differences in call density, noise levels, and the higher presence of multiples due to surface reflections compared to the Svalbard dataset (F1 = 0.877). Although the Svalbard-trained model already performs well, briefly fine-tuning the model on a small labeled Monaco–Italy subset could further enhance accuracy by adapting to these local acoustic conditions.

V. DATA PROCESSING PIPELINE FOR REAL-TIME DEPLOYMENT

One of the key challenges in deploying fin whale detection methods in real-time is efficient processing the large volumes of data collected by the interrogator units. For example, the Svalbard dataset is saved in HDF5 format, with each file containing 10 s of data. Each HDF5 file is approximately 420 MB, resulting in a daily data volume of roughly 3.7 TB. To handle this massive data size, an efficient batch processing pipeline is developed to process the data in real-time.

The pipeline processes 30 s of data with a 10-s overlap to ensure that calls at batch boundaries are not missed. This means it runs whenever two new HDF5 files become available. Therefore, the entire algorithm in Fig. 3 must process 30 s of data within 20 s, before the next batch is available. Figure 13 shows the average processing time for each method on the Svalbard dataset on a MacBook with an M2 Pro chip and 16 GB of RAM. The HT method takes approximately 17.8 s, while DBSCAN takes 21.4 s. The TM and YOLO methods take 17.8 s and 18.0 s, respectively. This indicates that, except for DBSCAN, all methods can be processed within the 20-s window, enabling real-time detection of fin whale calls.

The most time-consuming step is the preprocessing of the data, which includes loading and FK filtering. Regarding the core detection, YOLO is the fastest method, taking only 2.6 to process 30 s of data. The HT and TM methods are slightly slower at 2.9 and 3.0 s, respectively, while DBSCAN is the slowest, taking 5.5 s. The refinement step,

TABLE III. Detection performance of YOLO on Svalbard test data using column-wise versus image-wise transformation.

Method	P	R	F1	mF1
Image-wise	0.855	0.863	0.859	0.745
Column-wise	0.870	0.885	0.877	0.758

TABLE IV. Detection performance metrics for YOLO on the Monaco–Italy test set.

TP	FP	FN	P	R	F1	mF1
929	290	172	0.762	0.844	0.801	0.341

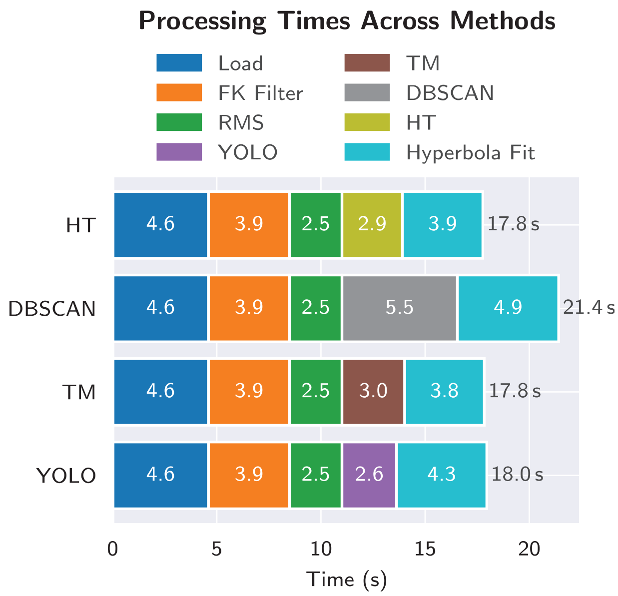

FIG. 13. Average processing time on 30-s batches of the Svalbard dataset. The HT method takes approximately 17.8 s, while DBSCAN takes 21.4 s. The TM and YOLO methods take 17.8 and 18.0 s, respectively. This indicates that, except for DBSCAN, all methods can be processed within the 20-s window, enabling real-time detection of fin whale calls.

which involves fitting hyperbolas, also contributes some to the processing time. The average processing time depends on the number of calls in the batch, as more calls require additional time to process. The processing times are expected to improve on the final deployment system, which is equipped with more powerful hardware. This enhancement will likely enable real-time deployment of the DBSCAN method as well.

The detected bounding boxes are assigned a unique *CallID*s. To prevent duplicate *CallID*s due to overlapping batches, the IoU score is calculated between the bounding boxes in the current batch and the previous batch. If the IoU score is greater than 0.5, the *CallID* of the previous batch is assigned to the bounding box in the current batch. If not, a new *CallID* is assigned. An example of the detection results stored in a PostgreSQL database is shown in Table V.

A dashboard is connected to the PostgreSQL database to visualize data in near real-time, allowing users to monitor the detection results (Fig. 14). The dashboard provides a user-friendly interface for exploring the detected calls, including their locations, times, and other relevant information. This enables researchers and conservationists to analyze the data effectively and make informed decisions based on the findings.

VI. CONCLUSION

The results of this study demonstrate that automated detection of fin whale calls using DAS data is both feasible and effective, particularly when using deep learning methods. YOLO achieves superior performance, with the highest precision, recall, and overall F1 scores compared to the other methods tested. Its success is attributed to its ability to learn complex patterns directly from labeled data, allowing it to robustly detect whale vocalizations even in noisy environments.

The post-refinement step of hyperbola fitting and LightGBM classifier further enhanced the precision of all methods, though it reduced recall slightly. This trade-off was generally acceptable, given the significant decrease in FPs. Methods like HT and DBSCAN showed varying levels of effectiveness, with DBSCAN particularly sensitive to noise, resulting in many FPs. TM followed by DBSCAN provided a good balance between precision and recall, but was still less flexible and effective compared to YOLO.

Importantly, YOLO demonstrated strong generalization capability by achieving good performance on the Monaco–Italy dataset without any additional fine-tuning. This suggests that models trained on one dataset can effectively be deployed in different locations, highlighting the practicality of this approach for real-time monitoring across various marine environments. Fine-tuning YOLO further for specific locations could likely enhance performance with relatively little additional effort.

An advantage of our approach is the capability to estimate whale positions by fitting hyperbolas to detected calls. While single-cable data allows distance estimation, the simultaneous use of two parallel cables, as shown by Rørstadbotnen *et al.* (2023), could enable precise localization of whales.

Despite the promising outcomes, several points remain for future improvement. First, the four methods we compared were selected to represent a wide range of detection paradigms. While this provided a broad performance overview, future work could focus on more detailed comparisons within specific algorithm families. For example, further studies might evaluate additional deep learning architectures like transformers or Faster R-CNN, which may offer improved detection under challenging acoustic conditions (Zhao *et al.*, 2024).

Second, we performed detection in the time–space domain. While this approach was effective for the fin whale vocalizations examined here, future work could analyze spectrogram representations or combine time–frequency techniques with spatial data to capture additional signal features. Such multi-domain strategies may further improve accuracy and robustness (Goestchel *et al.*, 2025; Bouffaut *et al.*, 2025).

TABLE V. An example of PostgreSQL data table after tracking.

Box ID	Call ID	Time Start	Distance Start	Time End	Distance End
1	1	2022-08-22 08:30:12	10	2022-08-22 08:30:20	30
2	2	2022-08-22 10:15:04	45	2022-08-22 10:15:10	60
3	2	2022-08-22 10:15:03	46	2022-08-22 10:15:11	59

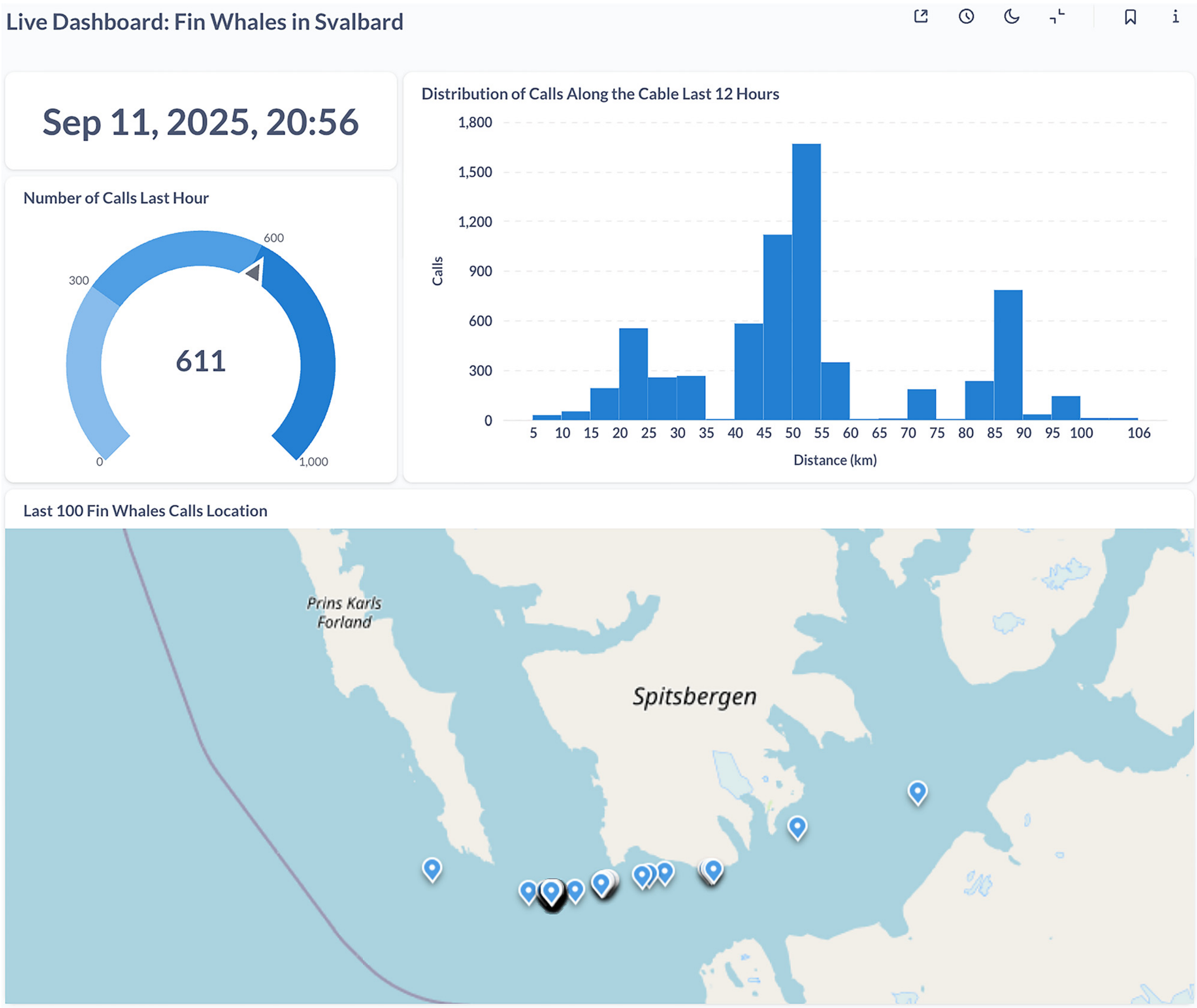

FIG. 14. An example of the dashboard connected to the PostgreSQL database. The dashboard provides a user-friendly interface for exploring the detected calls, including their locations, times, distribution, and other relevant information.

Third, although this study focused on fin whale vocalizations, the developed methodologies could be adapted to detect various other marine mammal calls or acoustic events such as vessel traffic or earthquakes. This would significantly broaden the applicability of DAS-based monitoring systems.

Overall, integrating DAS technology with advanced automated detection methods presents an important step towards scalable and continuous monitoring of marine mammals. Future research focusing on improved data efficiency, the incorporation of additional acoustic representations, and broader application to different acoustic signals will likely enhance detection capabilities and support marine conservation efforts effectively.

SUPPLEMENTARY MATERIAL

See supplementary material for the detailed derivation of Eqs. (3) and (6), as well as additional figures demonstrating the performance of the proposed methods on extra time windows.

ACKNOWLEDGMENTS

This work was supported by the Norwegian Research Council through the SFI Centre for Geophysical Forecasting (CGF) (Grant No. 309960) and by the EU Horizon 2020 program under the Marie Skłodowska-Curie grant agreement 101034240. We thank SIKT for providing the Svalbard DAS data. L.A. received support from the MARMOR project (ANR-21-ESRE-0020, France 2030), which also funded the DAS acquisition on the Monaco–Italy cable, with cable access enabled by Monaco Telecom. We are also grateful to Professor Cédric Richard and Professor André Ferrari at Côte d'Azur University for their valuable feedback and discussions.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The code used in this study is available at https://doi.org/10.5281/zenodo.17979619. Some of the processed Svalbard data are accessible at https://doi.org/10.18710/Q8OSON, while the remaining data cannot be shared due to confidentiality restrictions.

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